

Mafia FCIT20 : <https://t.me/FCIT20Male>

4 marks, section 1

MCQs, choose the correct answer:

1. the process subsystem is not responsible for which of the following :

- a- process creation**
- b- process scheduling**
- c- process termination**
- d- process memory allocation**

2. _____ memory technique is used to increase the degree of multiprogramming

- a- swapping**
- b- scheduling**
- c- dispatching**
- d- context switch**

3. when a user process calls an operating system call _____ occurs

- a- interrupt**
- b- trap**
- c- exception**
- d- none of the above**

4. the process longterm scheduler is responsible for managing effectively ?

- a- cpu**
- b- i/o devices**
- c- mother board**
- d- non-of-the above**

5. the scheduler for a batch system does not make use of the _____ to switch the running process from running to ready state

- a- input devices
- b- clock
- c- memory
- d- non-of-the above

6. the process short term scheduler _____

- a- is called frequently
- b- needs to decide very fast
- c- controls which process can enter the running state
- d- all of the above

7. the process dispatcher is responsible for _____

- a- selecting which process to be moved from ready state to running
- b- selecting which process to be waiting from waiting state to running
- c- save all cpu registers into pcb and load the cpu registers with pcb
- d. all of the above

8. the in round robin scheduling algorithm choosing the quantum too small will result on _____ scheduling algorithm is designed especially for time sharing systems

- a- better response time
- b- slow response time
- c- Answer A and better cpu utilization
- d- Answer A and lower cpu utilization

9. when a process uses blocking send system call in a message passing system the kernel puts the calling process in _____ state until a message is received

- a- new
- b- waiting
- c- running
- d- ready

10. each hardware device connected to a computer system is assigned a unique number to identify it when interrupts occurs the operating system uses this number to find out the address of the corresponding _____

- a- interrupt context switch
- b- interrupt handler routine
- c- interrupt vector
- d- dispatcher

11. a process enters the waiting queue when _____

- a- it finishes its time quantum
- b- it requests i/o operation
- c- interrupt occurs
- d- none of the above

12. what is not true with timesharing multitasking system ?

- a- operating system try to utilize the computer hardware resources efficiently**
- b- operating system gives each process the cpu for certain period of time**
- c- application couldn't know the time when they will get running**
- d- application could not cooperate together in executing a common task**

13. batch system cannot be used for _____

- a- applications that run one after the other**
- b- applications that don't allow sharing of computer resources with others**
- c- interactive application**
- d- none of the above**

14. which of the following scheduling algorithms must be non-preemptive?

- a- priority scheduling**
- b- multi levels feedback queues**
- c- shortest job first**
- d- first in first serve**

15. _____ is the number of processes that are completed per time unit

- a- cpu utilization**
- b- response time**
- c- turnaround time**
- d- throughput**

16. which processes scheduling algorithm gives the minimum average waiting time ?

a- priority scheduling

b- fcfs

c- sjf

d= none of the above

16 marks, section 2

Q9/ three processes a, b, c arrived in this order at the same time with the following cpu burst values

	CPU burst
A	9
B	5
C	7

Calculate the average waiting time & average turn around time by using FCFS & Round Robin (q=4) then decide which algorithm is better:

(نصيحه: مهم مهم جداً تدرك كل انواع الخوارزميات المستخدمة في ال

CPU Scheduling

لانه 100% بيجيك سؤال عنها بالاختبار وتعتبر درجات هديه)

Q1/ what is the role of the operating system, mention and explain only 2 roles:

Q2/ mention 5 process attributes that are stored in the process control block PCB data structure of any process ? explain the role of each one briefly :

Q3/ what is a context switch ? give one example when it occurs in user mode and one example in kernel mode :

Q4/ explain how the operating system scheduler uses a timer to prevent a process to run long period than it is allowed or deserved ?

Q5/ explain the difference of the" kernel mode" and "user mode" during the execution of a process :

Q6/ mention the role of the process "text segment" and "data segment":

Q7/ when multiple processes need to cooperate there is a choice between shared memory and inter process communication (ipc)

states 3 advantages of each mechanisms &

make sure to clarify the role of the operating system in each:

Q8/ mention 2 goals of good scheduling algorithm and explain their conflicts to achieve both of them ?